

Scenario: "PustokBazar" Online Bookstore Database

You are the database developer for an online bookstore. The system tracks **customers**, **books**, **orders**, and the **items within each order**. Use the schema and sample data below to answer all questions. Write your queries in standard SQL (MySQL syntax).

Table 1: Customers

CustomerID (PK)	Name	City	JoinDate
1	Rahim Uddin	Dhaka	2024-01-10
2	Karim Ahmed	Chattogram	2024-03-22

Table 2: Books

BookID (PK)	Title	Category	Price	Stock
101	Database Systems	Technology	850	20
102	Bangla Kobita Shomogro	Literature	400	5
103	Clean Code	Technology	950	0
104	Economics of Bangladesh	Business	600	15
105	Networking Basics	Technology	700	8

Table 3: Orders

OrderID (PK)	CustomerID (FK)	OrderDate	Status
1001	1	2026-06-01	Delivered
1002	2	2026-06-05	Delivered

Table 4: OrderItems

OrderItemID (PK)	OrderID (FK)	BookID (FK)	Quantity
1	1001	101	1

2	1001	105	2
3	1002	102	1

Q1. Write a query to display the **Title**, **Category**, and **Price** of all books priced above **600**, sorted by **Price** in descending order. (10 marks)

Q2. Write a query to display the number of customers who joined from each **City**, showing only cities with **more than 1** customer. (10 marks)

Q3. Write a query using a subquery to find the **Title** of the book(s) with the **highest price** in the **Books** table (without using **ORDER BY ... LIMIT**). (10 marks)

Q4. Write a query using a **correlated subquery** to list all customers (**Name**) who have placed **at least one order** with a **Status** of '**Delivered**'. (10 marks)

Q5. Write a query using **INNER JOIN** to display the **OrderID**, customer **Name**, and **OrderDate** for all orders placed by customers from '**Dhaka**'. (10 marks)

Q6. Write a query using a **LEFT JOIN** to list every book's **Title** along with the total **Quantity** ordered across all orders — including books that have **never been ordered** (their total should show as 0 or NULL). (10 marks)

Q7. Write a query joining all four tables (**Customers**, **Orders**, **OrderItems**, **Books**) to display: customer **Name**, **OrderID**, book **Title**, **Quantity**, and the line total (**Quantity × Price**), for orders with **Status = 'Delivered'**. (10 marks)

Q8. Write a **Stored Procedure** named **GetOrdersByCustomer** that accepts a **CustomerID** as an input parameter and returns all orders (**OrderID**, **OrderDate**, **Status**) placed by that customer. (10 marks)

Q9. Write a **Stored Procedure** named **UpdateStock** that accepts a **BookID** and a **Quantity** as input parameters, and reduces that book's **Stock** by the given quantity — but only if enough stock is available (otherwise it should not update and should indicate insufficient stock). (10 marks)

Q10. Create a **View** named **DeliveredOrderSummary** that shows **CustomerName**, **OrderID**, **OrderDate**, and the total order value (sum of **Quantity × Price** for all items in that order), limited to orders with **Status = 'Delivered'**. Then write a query that uses this view to find the customer with the **highest total single-order value**. (10 marks)